THE AUGMENTED CECH COMPLEX OF A MODULE AND ITS EXACTNESS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the resolution input is Lemma lemma-simplicial-resolution-augmentation-modules (in-source stub); adjointness of pullback and pushforward is the transport mechanism.

Editorial note. The label lemma-augmentation-cech-complex-modules names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Simplicial resolution of modules. Same unit, Lemma lemma-simplicial-resolution-augmentation-modules. Provides the exact complex ... -> g_{n!}(a_n^*G) -> ... -> a^*G -> 0 that Hom(-, F) is applied to.

Source reference. Original reference 0D7C: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 2326–2326. Prerequisite pointer invoked at proof line(s) 2386; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *With notation as above. For an $\mathcal{O}$-module $\mathcal{F}$ on $\mathcal{C}_{total}$ there is a canonical complex*

$$0 \to a_*\mathcal{F} \to a_{0,*}\mathcal{F}_0 \to a_{1,*}\mathcal{F}_1 \to a_{2,*}\mathcal{F}_2 \to \ldots$$

of $\mathcal{O}_\mathcal{D}$-modules which is exact in degrees $-1, 0$. If $\mathcal{F}$ is an injective $\mathcal{O}$-module, then the complex is exact in all degrees and remains exact on applying the functor $\mathrm{Hom}_{\mathcal{O}_\mathcal{D}}(\mathcal{G}, -)$ for any $\mathcal{O}_\mathcal{D}$-module $\mathcal{G}$.

Proof. To construct the complex, by the Yoneda lemma, it suffices for any $\mathcal{O}_\mathcal{D}$-modules $\mathcal{G}$ on $\mathcal{D}$ to construct a complex

$$0 \to \mathrm{Hom}_{\mathcal{O}_\mathcal{D}}(\mathcal{G}, a_*\mathcal{F}) \to \mathrm{Hom}_{\mathcal{O}_\mathcal{D}}(\mathcal{G}, a_{0,*}\mathcal{F}_0) \to \mathrm{Hom}_{\mathcal{O}_\mathcal{D}}(\mathcal{G}, a_{1,*}\mathcal{F}_1) \to \ldots$$

functorially in $\mathcal{G}$. To do this apply $\mathrm{Hom}_\mathcal{O}(-, \mathcal{F})$ to the exact complex of Lemma 0D7C and use adjointness of pullback and pushforward. The exactness properties in degrees $-1, 0$ follow from the construction as $\mathrm{Hom}_\mathcal{O}(-, \mathcal{F})$ is left exact. If $\mathcal{F}$ is an injective $\mathcal{O}$-module, then the complex is exact because $\mathrm{Hom}_\mathcal{O}(-, \mathcal{F})$ is exact. □

Date: stacks-project:spaces-simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.